CAS SciFinder

Task History

Initiating Search

November 4, 2025, 11:11 AM

Search:

CAS Lexicon Search:

Negishi coupling reaction - Preferred Concept

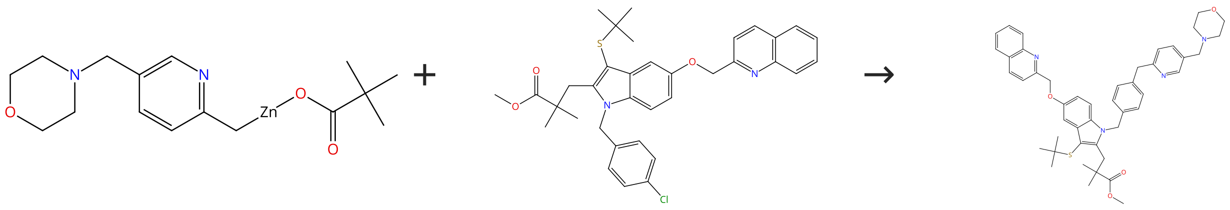
Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,232)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (2,632) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal**	**Reactions**	View Results

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (14)

31-176-CAS-16294017 Steps: **1** Yield: **100%**

1.1 **Catalysts:** (*SP*-4-3)-[2'-(Amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κ*O*)palladium
Solvents: Tetrahydrofuran; 18 h, 50 °C

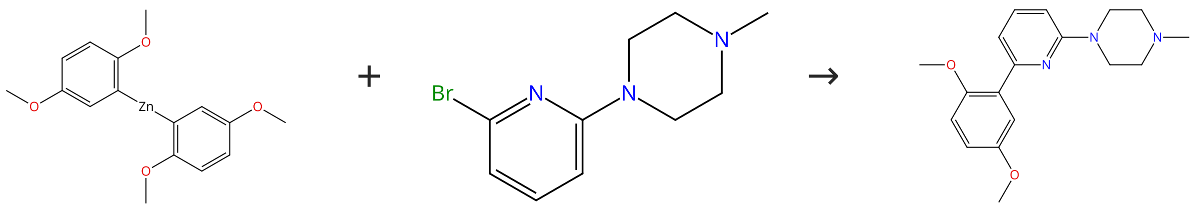
Experimental Protocols

Synthesis of Complex Druglike Molecules by the Use of Highly Functionalized Bench-Stable Organozinc Reagents

By: Greshock, Thomas J.; et al

Angewandte Chemie, International Edition (2016), 55(44), 13714-13718.

Scheme 2 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (57) Supplier (1)

31-176-CAS-8636720 Steps: **1** Yield: **100%**

1.1 **Solvents:** Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; rt → 100 °C

1.2 **Catalysts:** (*SP*-4-1)-Dichlorobis[1-(dicyclohexylphosphino-κ*P*)piperidine]palladium
Solvents: Toluene; 15 min, 100 °C; 100 °C → rt

1.3 **Reagents:** Ethylenediaminetetraacetic acid, Sodium hydroxide
Solvents: Water; rt

Experimental Protocols

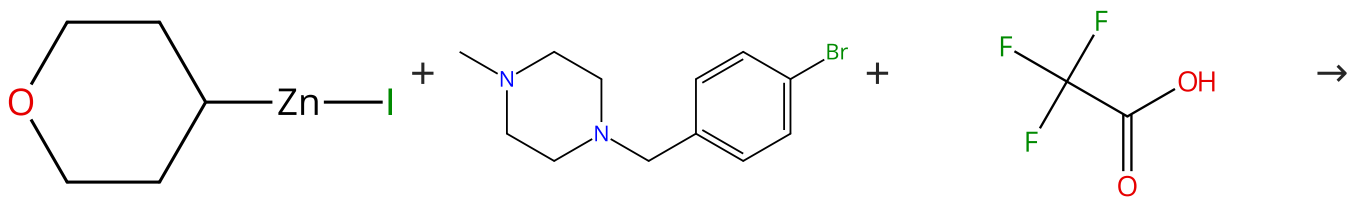
Access to 2-Aminopyridines - Compounds of Great Biological and Chemical Significance

By: Bolliger, Jeanne L.; et al

Advanced Synthesis & Catalysis (2011), 353(6), 945-954.

Scheme 3 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (11)

Suppliers (54)

Suppliers (192)

Multi-component structure image available in CAS SciFinder

31-176-CAS-21867514

Steps: **1** Yield: **100%**

1.1 **Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-4,5-dichloro-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Tetrahydrofuran; overnight, rt

1.2 -

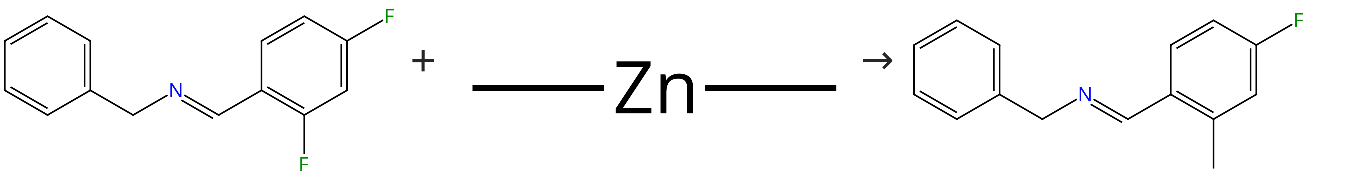
Expanding the Medicinal Chemist Toolbox: Comparing Seven C(sp^2)-C(sp^3) Cross-Coupling Methods by Library Synthesis

By: Dombrowski, Amanda W.; et al

ACS Medicinal Chemistry Letters (2020), 11(4), 597-604.

Scheme 4 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (31)

31-176-CAS-1767861

Steps: **1** Yield: **100%**

1.1 **Catalysts:** Dichlorobis(triethylphosphine)nickel
Solvents: Acetonitrile; 24 h, 60 °C

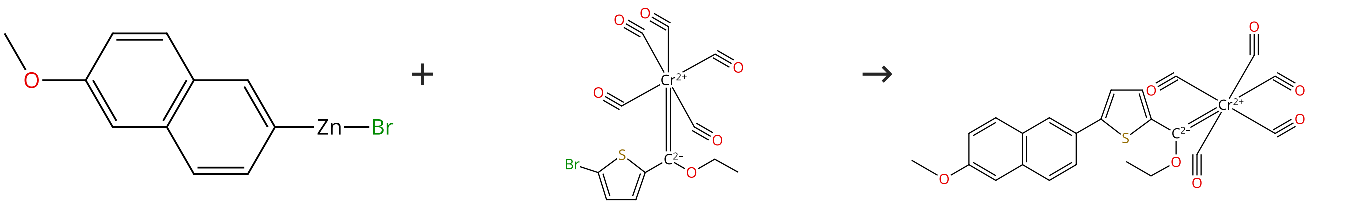
Nickel-Catalyzed Csp^2-Csp^3 Bond Formation by Carbon-Fluorine Activation

By: Sun, Alex D.; et al

Chemistry - A European Journal (2014), 20(11), 3162-3168.

Scheme 5 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (8)

31-176-CAS-10198669 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Tetrakis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; rt → -78 °C; 3 h, rt

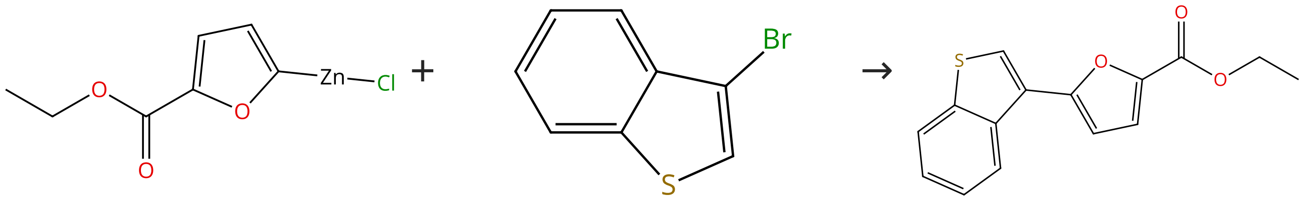
Experimental Protocols

Negishi Cross-Coupling Reaction as a Simple and Efficient Route to Functionalized Amino and Alkoxy Carbene Complexes of Chromium, Molybdenum, and Tungsten

By: Tobrman, Tomas; et al

Organometallics (2014), 33(22), 6593-6603.

Scheme 6 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (72) Supplier (1)

31-176-CAS-2761074 Steps: **1** Yield: **100%**

1.1 **Reagents:** Zinc chloride
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Tetrahydrofuran; 2 min, rt

1.2 **Solvents:** Tetrahydrofuran; 24 h, 23 °C

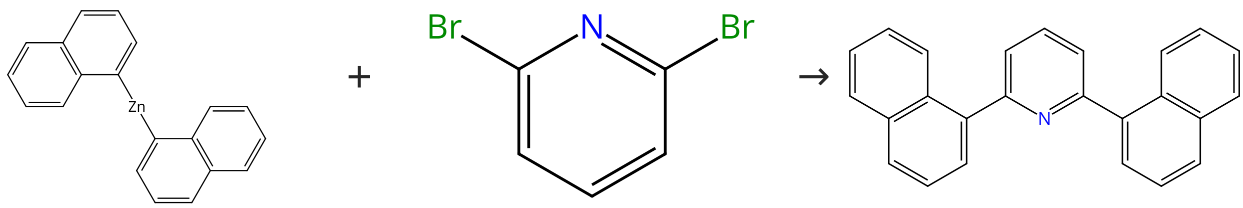
Experimental Protocols

Pd-PEPPSI-IPent: Low-Temperature Negishi Cross-Coupling for the Preparation of Highly Functionalized, Tetra-ortho-Substituted Biaryls

By: Calimsiz, Selcuk; et al

Angewandte Chemie, International Edition (2010), 49(11), 2014-2017, S2014/1-S2014/76.

Scheme 7 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (85) Suppliers (4)

31-176-CAS-2496719 Steps: **1** Yield: **100%**

1.1 **Solvents:** Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; rt → 100 °C

1.2 **Catalysts:** (*SP*-4-1)-Dichlorobis[1-(dicyclohexylphosphino-κ*P*)piperidine]palladium
Solvents: Tetrahydrofuran; 30 min, 100 °C; 100 °C → rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; rt

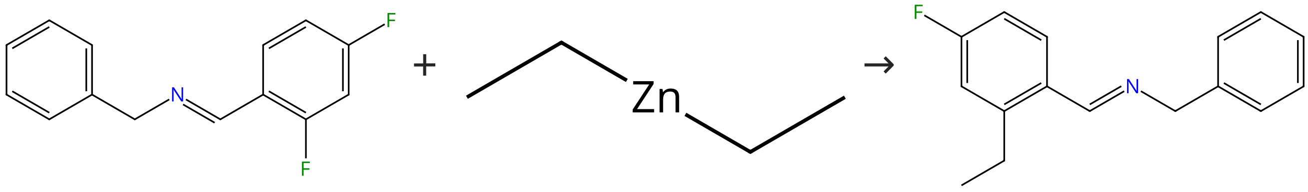
Experimental Protocols

$Pd(Cl)_2\{P(NC_5H_{10})(C_6H_{11})_2\}_2$ - A highly effective and extremely versatile palladium-based negishi catalyst that efficiently and reliably operates at low catalyst loadings

By: Bolliger, Jeanne L.; et al

Chemistry - A European Journal (2010), 16(36), 11072-11081, S11072/1-S11072/22.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (54)

31-176-CAS-3901173 Steps: **1** Yield: **100%**

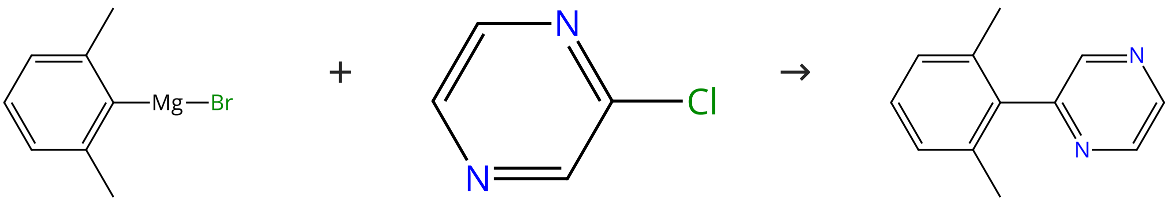
1.1 **Catalysts:** Dichlorobis(triethylphosphine)nickel
Solvents: Acetonitrile; 24 h, 60 °C

Nickel-Catalyzed Csp^2-Csp^3 Bond Formation by Carbon-Fluorine Activation

By: Sun, Alex D.; et al

Chemistry - A European Journal (2014), 20(11), 3162-3168.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (19) Suppliers (89) Suppliers (9)

31-176-CAS-670578 Steps: **1** Yield: **100%**

1.1 **Reagents:** Zinc chloride
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*) palladium
Solvents: Tetrahydrofuran; 10 min, rt

1.2 20 min, rt

1.3 **Solvents:** *N*-Methyl-2-pyrrolidone; 24 h, 23 °C

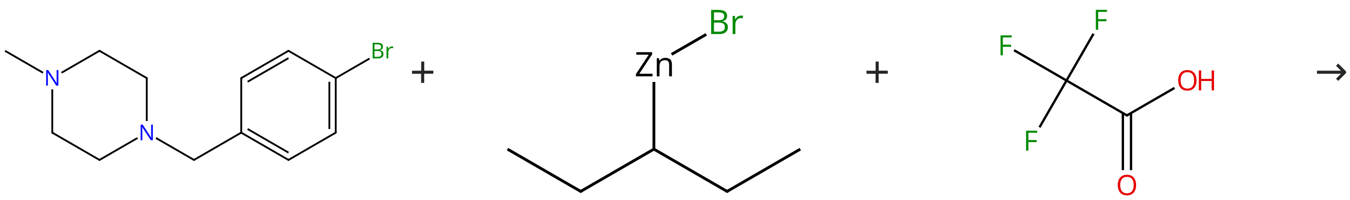
Experimental Protocols

Pd-PEPPSI-IPent: Low-Temperature Negishi Cross-Coupling for the Preparation of Highly Functionalized, Tetra-ortho-Substituted Biaryls

By: Calimsiz, Selcuk; et al

Angewandte Chemie, International Edition (2010), 49(11), 2014-2017, S2014/1-S2014/76.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (54) Suppliers (22) Suppliers (192)

Multi-component structure image available in CAS SciFinder

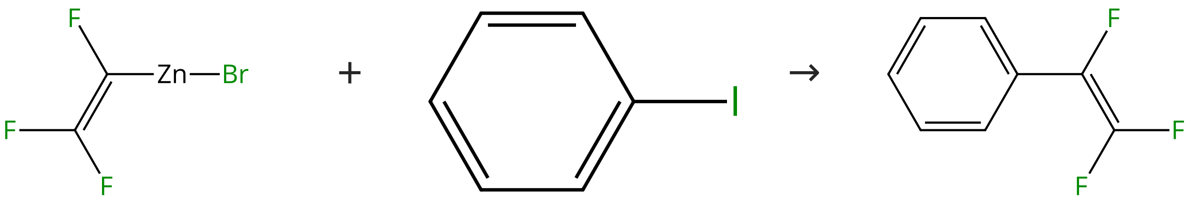
31-176-CAS-21867600	Steps: **1** Yield: **100%**
1.1	**Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-4,5-dichloro-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium **Solvents:** Tetrahydrofuran; overnight, rt
1.2	-

Expanding the Medicinal Chemist Toolbox: Comparing Seven $C(sp^2)$-$C(sp^3)$ Cross-Coupling Methods by Library Synthesis

By: Dombrowski, Amanda W.; et al

ACS Medicinal Chemistry Letters (2020), 11(4), 597-604.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2) Suppliers (95) Suppliers (21)

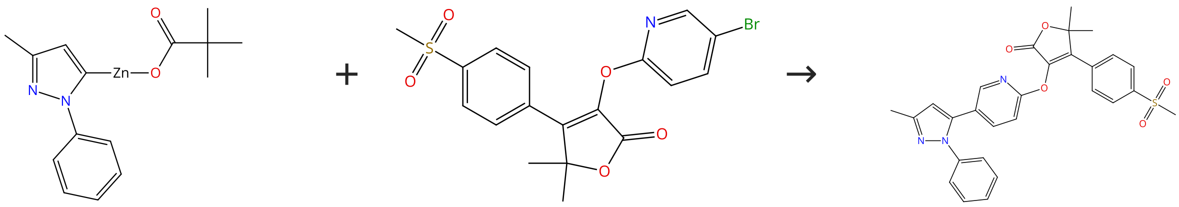
31-176-CAS-6679654	Steps: **1** Yield: **100%**
1.1	**Catalysts:** Tetrakis(triphenylphosphine)palladium **Solvents:** Dimethylformamide; 12 h, rt

The stereoselective synthesis of (Z)-HFC=CFZnI and stereospecific preparation of (E)-1,2-difluorostyrenes from (Z)-HFC=CFZnI via an unusual $Pd(PPh_3)_4$-Cu(I)Br co-catalysis approach or (Z)-HFC=$CFSnBu_3$

By: Liu, Qibo; et al

Journal of Fluorine Chemistry (2011), 132(2), 78-87.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (8)

31-176-CAS-16293993	Steps: **1** Yield: **100%**
1.1	**Catalysts:** (*SP*-4-4)-[2'-(Amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*][4-(diphenylphosphino-κ*P*)-6-(diphenylphosphino)-10*H*-phenoxazine](methanesulfonato-κ*O*)palladium **Solvents:** Tetrahydrofuran; 18 h, 50 °C

Experimental Protocols

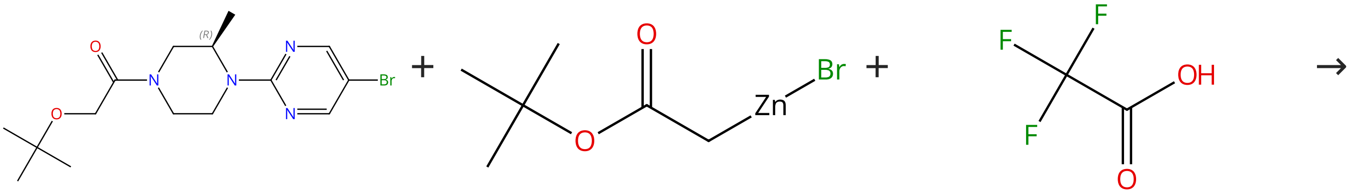
Synthesis of Complex Druglike Molecules by the Use of Highly Functionalized Bench-Stable Organozinc Reagents

By: Greshock, Thomas J.; et al

Angewandte Chemie, International Edition (2016), 55(44), 13714-13718.

Scheme 13 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Supplier (1)

Suppliers (20)

Suppliers (192)

Multi-component structure image available in CAS SciFinder

31-176-CAS-21867593

Steps: **1** Yield: **100%**

1.1 **Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-4,5-dichloro-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Tetrahydrofuran; overnight, rt

1.2 -

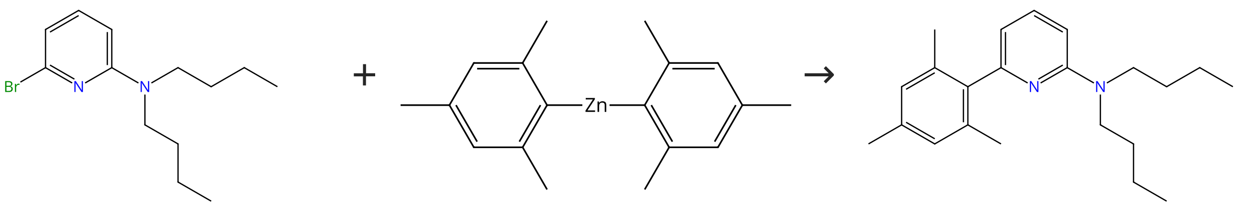
Expanding the Medicinal Chemist Toolbox: Comparing Seven C(sp^2)-C(sp^3) Cross-Coupling Methods by Library Synthesis

By: Dombrowski, Amanda W.; et al

ACS Medicinal Chemistry Letters (2020), 11(4), 597-604.

Scheme 14 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (7)

Supplier (1)

Supplier (1)

31-176-CAS-8692003

Steps: **1** Yield: **100%**

1.1 **Solvents:** Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; rt → 100 °C

1.2 **Catalysts:** (*SP*-4-1)-Dichlorobis[1-(dicyclohexylphosphino-κ*P*)piperidine]palladium
Solvents: Toluene; 15 min, 100 °C; 100 °C → rt

1.3 **Reagents:** Ethylenediaminetetraacetic acid, Sodium hydroxide
Solvents: Water; rt

Experimental Protocols

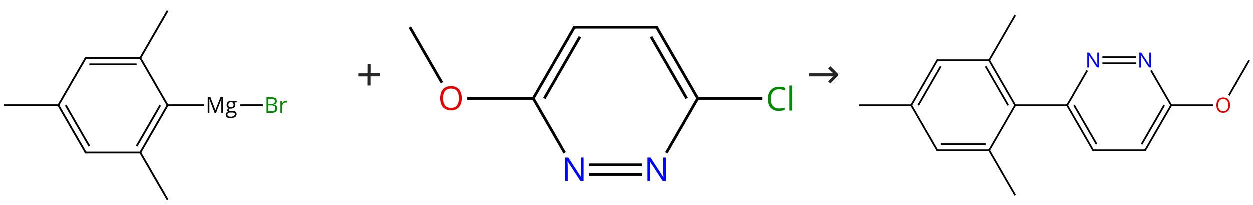
Access to 2-Aminopyridines - Compounds of Great Biological and Chemical Significance

By: Bolliger, Jeanne L.; et al

Advanced Synthesis & Catalysis (2011), 353(6), 945-954.

Scheme 15 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (27)

Suppliers (90)

Supplier (1)

31-176-CAS-14060028 Steps: **1** Yield: **100%**

1.1 **Reagents:** Zinc chloride
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Tetrahydrofuran; 10 min, rt

1.2 20 min, rt

1.3 **Solvents:** *N*-Methyl-2-pyrrolidone; 24 h, 23 °C

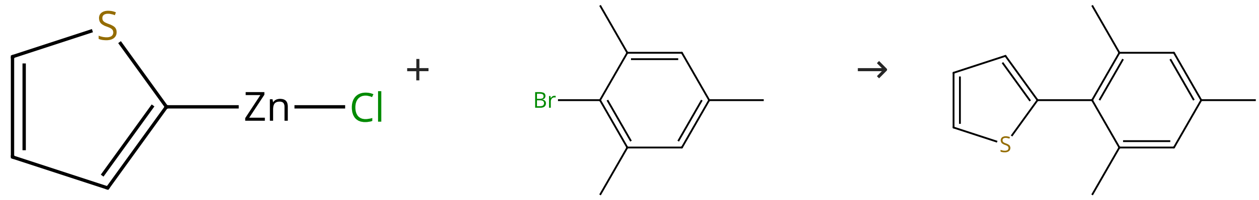
Experimental Protocols

Pd-PEPPSI-IPent: Low-Temperature Negishi Cross-Coupling for the Preparation of Highly Functionalized, Tetra-ortho-Substituted Biaryls

By: Calimsiz, Selcuk; et al

Angewandte Chemie, International Edition (2010), 49(11), 2014-2017, S2014/1-S2014/76.

Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (3) Suppliers (82) Suppliers (3)

31-176-CAS-4605097 Steps: **1** Yield: **100%**

1.1 **Reagents:** Zinc chloride
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Tetrahydrofuran; 2 min, rt

1.2 **Solvents:** Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; 24 h, 23 °C

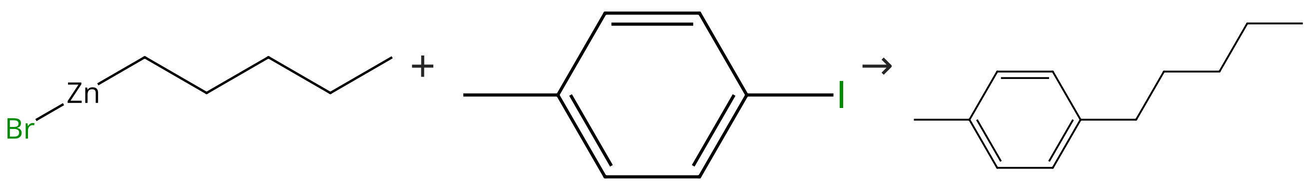
Experimental Protocols

Pd-PEPPSI-IPent: Low-Temperature Negishi Cross-Coupling for the Preparation of Highly Functionalized, Tetra-ortho-Substituted Biaryls

By: Calimsiz, Selcuk; et al

Angewandte Chemie, International Edition (2010), 49(11), 2014-2017, S2014/1-S2014/76.

Scheme 17 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (23) Suppliers (81) Suppliers (10)

31-176-CAS-16549202 Steps: **1** Yield: **100%**

1.1 **Catalysts:** Nickel, (2,2'-bipyridine-κN^1,κ$N^{1'}$)bromo(2,4,6-trimethylphenyl)-, (*SP*-4-2)-
Solvents: Tetrahydrofuran; 12 h, rt

1.2 **Solvents:** Water; rt

Experimental Protocols

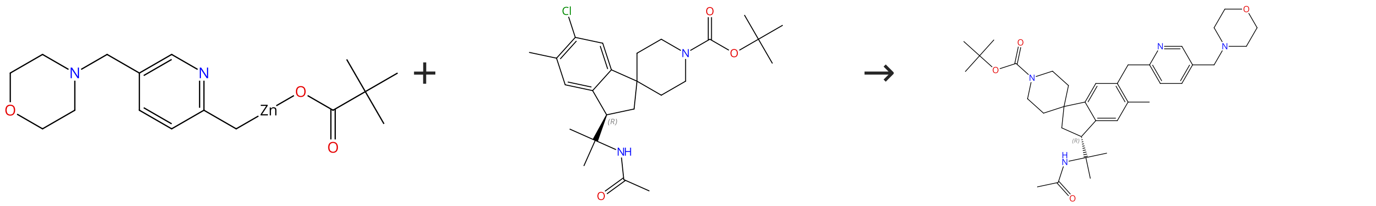
Unsymmetrical N-Aryl-1-(pyridin-2-yl)methanimine Ligands in Organonickel(II) Complexes: More Than a Blend of 2, 2'-Bipyridine and N,N-Diaryl-α-diimines?

By: Biewer, Christian; et al

Inorganic Chemistry (2016), 55(24), 12716-12727.

Scheme 18 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Absolute stereochemistry shown

Suppliers (3)

31-176-CAS-16294015	Steps: **1** Yield: **100%**
1.1	**Catalysts:** (*SP*-4-3)-[2'-(Amino-κ*N*)[1,1'-biphenyl]-2-yl-κ*C*][dicyclohexyl[2',4',6'-tris(1-methylethyl)[1,1'-biphenyl]-2-yl]phosphine](methanesulfonato-κ*O*)palladium **Solvents:** Tetrahydrofuran; 18 h, 50 °C

Experimental Protocols

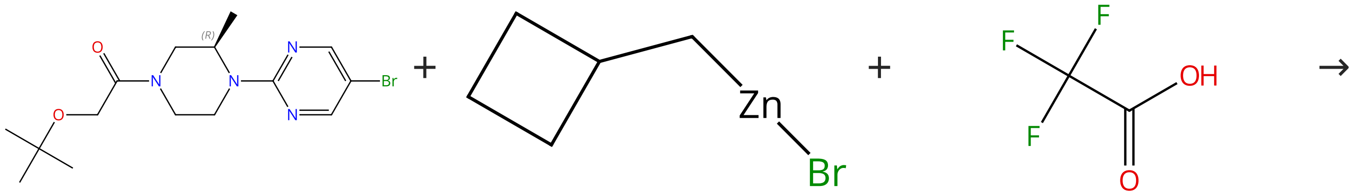
Synthesis of Complex Druglike Molecules by the Use of Highly Functionalized Bench-Stable Organozinc Reagents

By: Greshock, Thomas J.; et al

Angewandte Chemie, International Edition (2016), 55(44), 13714-13718.

Scheme 19 (1 Reaction)

Steps: **1** Yield: **100%**



Absolute stereochemistry shown

Supplier (1)

Suppliers (11)

Suppliers (192)

Multi-component structure image available in CAS SciFinder

31-176-CAS-21867542	Steps: **1** Yield: **100%**
1.1	**Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-4,5-dichloro-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium **Solvents:** Tetrahydrofuran; overnight, rt
1.2	-

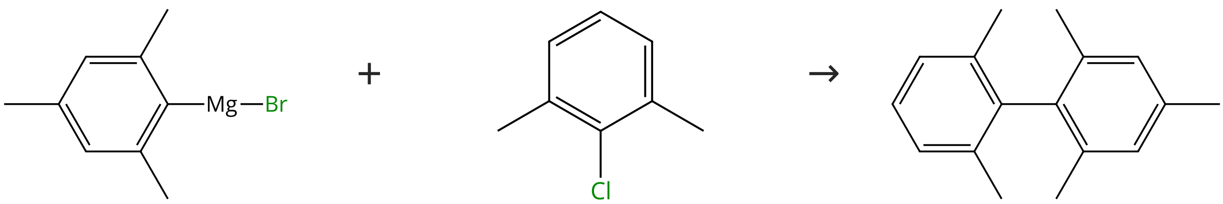
Expanding the Medicinal Chemist Toolbox: Comparing Seven C(sp^2)-C(sp^3) Cross-Coupling Methods by Library Synthesis

By: Dombrowski, Amanda W.; et al

ACS Medicinal Chemistry Letters (2020), 11(4), 597-604.

Scheme 20 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (27)

Suppliers (74)

Suppliers (11)

31-176-CAS-8453136 Steps: **1** Yield: **100%**

1.1 **Reagents:** Zinc chloride
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*) palladium
Solvents: Tetrahydrofuran; 10 min, rt

1.2 20 min, rt

1.3 **Solvents:** *N*-Methyl-2-pyrrolidone; 8 h, 23 °C

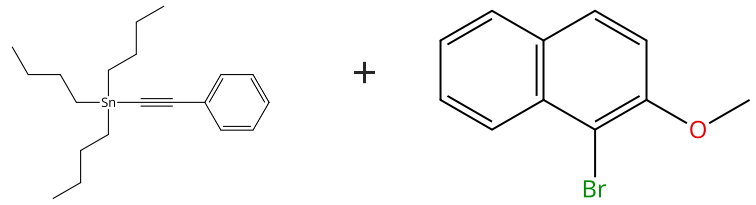
Experimental Protocols

Pd-PEPPSI-IPent: Low-Temperature Negishi Cross-Coupling for the Preparation of Highly Functionalized, Tetra-ortho-Substituted Biaryls

By: Calimsiz, Selcuk; et al

Angewandte Chemie, International Edition (2010), 49(11), 2014-2017, S2014/1-S2014/76.

Scheme 21 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (30) Suppliers (75) Suppliers (3)

31-130-CAS-254862 Steps: **1** Yield: **100%**

1.1 **Reagents:** Dabco, Sodium chloride, Poly(oxy-1,2-ethanediyl), α-[4-[[3,4-dihydro-2,5,7,8-tetramethyl-2-(4,8,12-trimethyltridecyl)-2*H*-1-benzopyran-6-yl]oxy]-1,4-dioxobutyl]-ω-methoxy-
Catalysts: Bis(tri-*tert*-butylphosphine)palladium
Solvents: Water; 4 h, rt

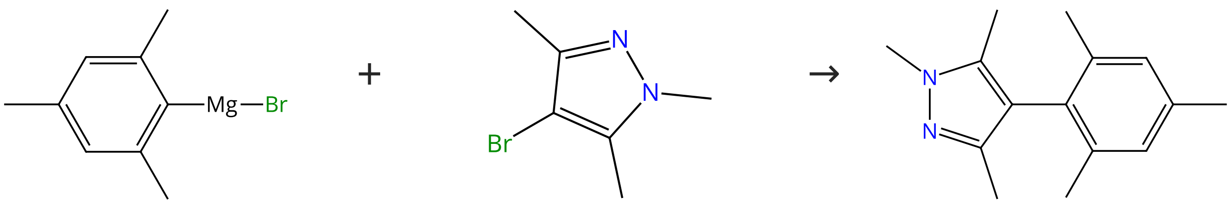
Experimental Protocols

"Nok": A Phytosterol-Based Amphiphile Enabling Transition-Metal-Catalyzed Couplings in Water at Room Temperature

By: Klumphu, Piyatida; et al

Journal of Organic Chemistry (2014), 79(3), 888-900.

Scheme 22 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (27) Suppliers (82) Supplier (1)

31-176-CAS-11940700 Steps: **1** Yield: **100%**

1.1 **Reagents:** Zinc chloride
Catalysts: (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*) palladium
Solvents: Tetrahydrofuran; 10 min, rt

1.2 20 min, rt

1.3 **Solvents:** *N*-Methyl-2-pyrrolidone; 24 h, 23 °C

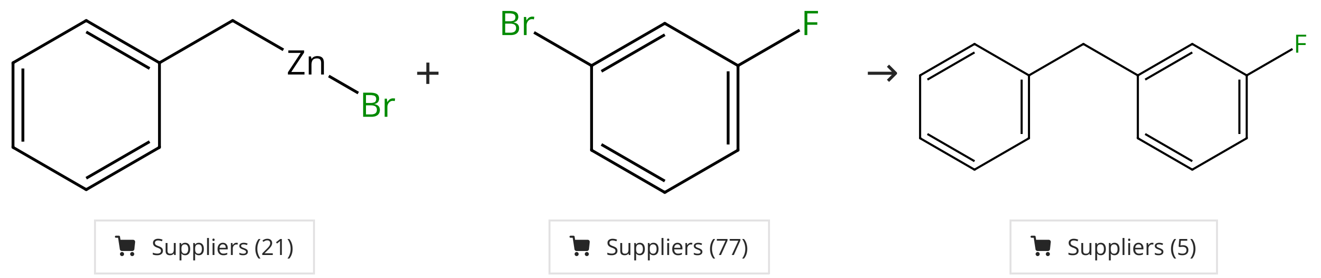
Experimental Protocols

Pd-PEPPSI-IPent: Low-Temperature Negishi Cross-Coupling for the Preparation of Highly Functionalized, Tetra-ortho-Substituted Biaryls

By: Calimsiz, Selcuk; et al

Angewandte Chemie, International Edition (2010), 49(11), 2014-2017, S2014/1-S2014/76.

Scheme 23 (1 Reaction) Steps: **1** Yield: **99%**



31-176-CAS-20748295 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-3)-Bromophenylbis(triphenylphosphine) palladium
Solvents: Tetrahydrofuran; rt; 4 h, 50 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water

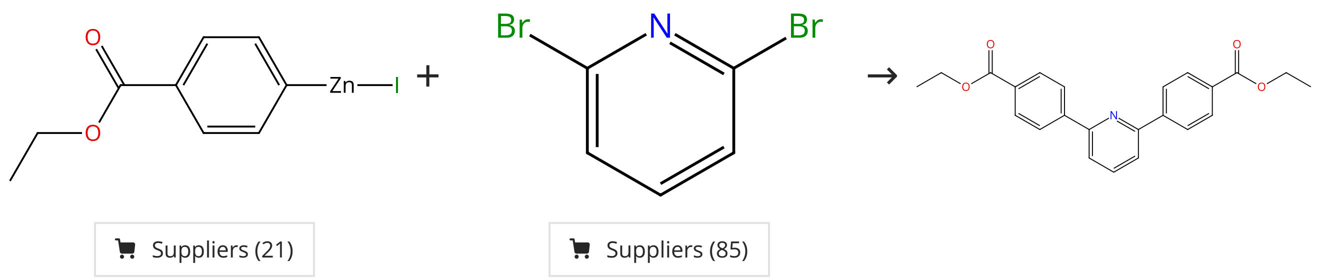
Experimental Protocols

POxAP Precatalysts and the Negishi Cross-Coupling Reaction

By: Tang, Shuang-Qi; et al

Synthesis (2020), 52(1), 51-59.

Scheme 24 (1 Reaction) Steps: **1** Yield: **99%**



31-176-CAS-10762743 Steps: **1** Yield: **99%**

1.1 **Solvents:** Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; rt → 100 °C

1.2 **Catalysts:** (*SP*-4-1)-Dichlorobis[1-(dicyclohexylphosphino-κ*P*) piperidine]palladium
Solvents: Tetrahydrofuran; 30 min, 100 °C; 100 °C → rt

1.3 **Reagents:** Hydrochloric acid
Solvents: Water; rt

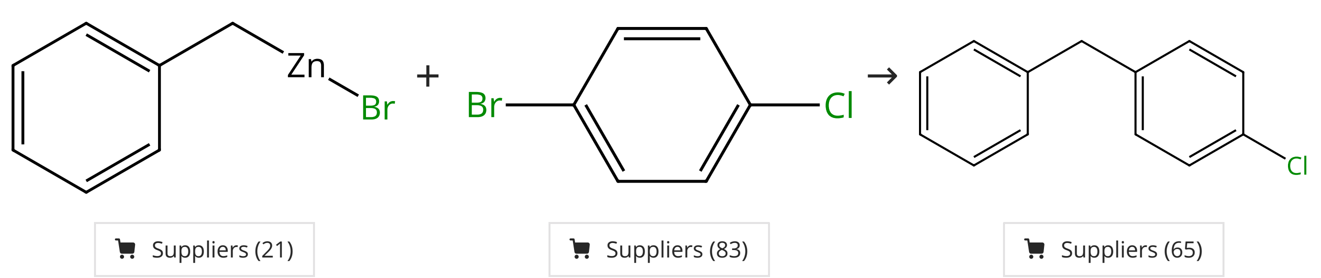
Experimental Protocols

$Pd(Cl)_2\{P(NC_5H_{10})(C_6H_{11})_2\}_2$ - A highly effective and extremely versatile palladium-based negishi catalyst that efficiently and reliably operates at low catalyst loadings

By: Bolliger, Jeanne L.; et al

Chemistry - A European Journal (2010), 16(36), 11072-11081, S11072/1-S11072/22.

Scheme 25 (1 Reaction) Steps: **1** Yield: **99%**



31-176-CAS-10149235 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-4)-[(Benzoic acid-κ*O*) 2-[3,3,3-trifluoro-3-(oxo-κ*O*)-1-(trifluoromethyl)butyl]hydrazidato(2-)-κN^2](triphenyl phosphine)nickel
Solvents: Tetrahydrofuran; 24 h, 70 °C

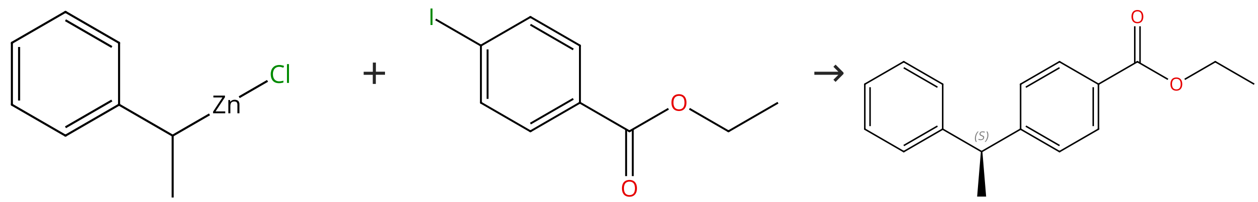
Experimental Protocols

Synthesis of Ni(II) complexes with unsymmetric [O,N,O']-pincer ligands and their use as precatalysts in carbon-carbon bond formations to access diarylmethanes

By: Someya, Chika I.; et al

Inorganica Chimica Acta (2014), 421, 136-144.

Scheme 26 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (6) Suppliers (83)

Absolute stereochemistry shown, Rotation (-)

31-614-CAS-42728039 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Palladium diacetate, 1,3-Dioxolo[4,5-*e*][1,3,2]dioxaphosphepin, 4,4,8,8-tetrakis[4-(1,1-dimethylethyl)phenyl]tetrahydro-2,2-dimethyl-6-[[(1*S*,2*R*,5*S*)-5-methyl-2-(1-methylethyl)cyclohexyl]oxy]-, (3a*S*,8a*S*)-
Solvents: *tert*-Butyl methyl ether; 30 min, rt

1.2 **Reagents:** Zinc chloride
Solvents: 2-Methyltetrahydrofuran, *tert*-Butyl methyl ether; 12 h, 23 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

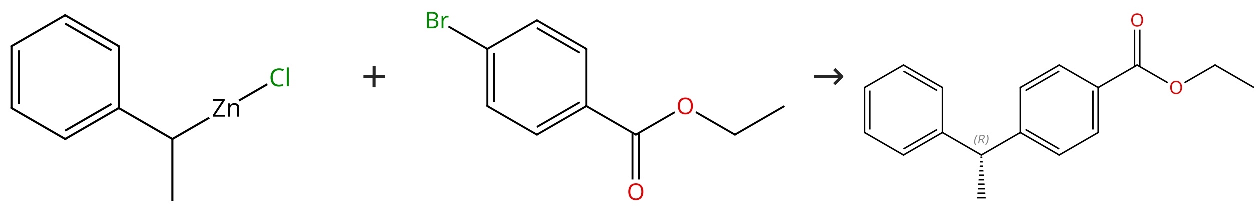
Experimental Protocols

Enantioconvergent Negishi Cross-Couplings of Racemic Secondary Organozinc Reagents to Access Privileged Scaffolds: A Combined Experimental and Theoretical Study

By: Preinfalk, Alexander; et al

Angewandte Chemie, International Edition (2024), 63(50), e202414868.

Scheme 27 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (6) Suppliers (81)

Absolute stereochemistry shown

31-614-CAS-42728025 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Di-μ-chlorobis[(1,2,3-η)-1-phenyl-2-propen-1-yl]dipalladium, 1,3-Dioxolo[4,5-*e*][1,3,2]dioxaphosphepin, 4,4,8,8-tetrakis[4-(1,1-dimethylethyl)phenyl]tetrahydro-2,2-dimethyl-6-[[(1*R*,2*S*,5*R*)-5-methyl-2-(1-methyl-1-phenylethyl)cyclohexyl]oxy]-, (3a*R*,8a*R*)-
Solvents: *tert*-Butyl methyl ether; 30 min, rt

1.2 **Reagents:** Zinc chloride
Solvents: 2-Methyltetrahydrofuran, *tert*-Butyl methyl ether; 12 h, 21 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

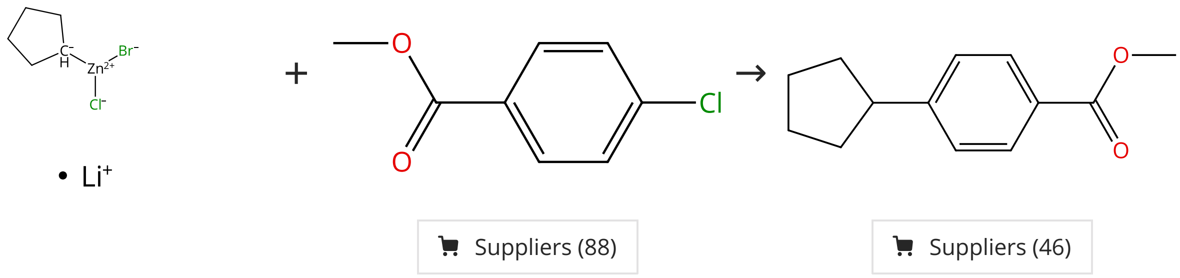
Enantioconvergent Negishi Cross-Couplings of Racemic Secondary Organozinc Reagents to Access Privileged Scaffolds: A Combined Experimental and Theoretical Study

By: Preinfalk, Alexander; et al

Angewandte Chemie, International Edition (2024), 63(50), e202414868.

Scheme 28 (1 Reaction)

Steps: **1** Yield: **99%**



31-176-CAS-14597323

Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-1)-[7,9-Bis[2,6-bis(1-methylethyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: 1,4-Dioxane; 0 °C; 30 min, rt

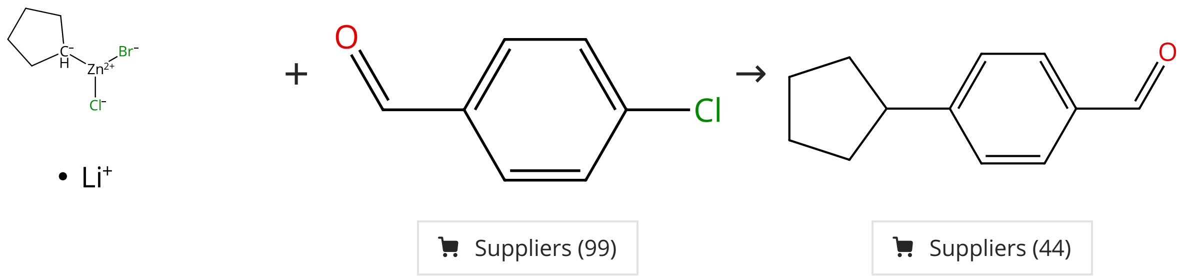
Mild Negishi Cross-Coupling Reactions Catalyzed by Acenaphthoimidazolylidene Palladium Complexes at Low Catalyst Loadings

By: Liu, Zelong; et al

Journal of Organic Chemistry (2013), 78(15), 7436-7444.

Scheme 29 (1 Reaction)

Steps: **1** Yield: **99%**



31-176-CAS-7275770

Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-1)-[7,9-Bis[2,6-bis(1-methylethyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: 1,4-Dioxane; 0 °C; 0.5 h, rt

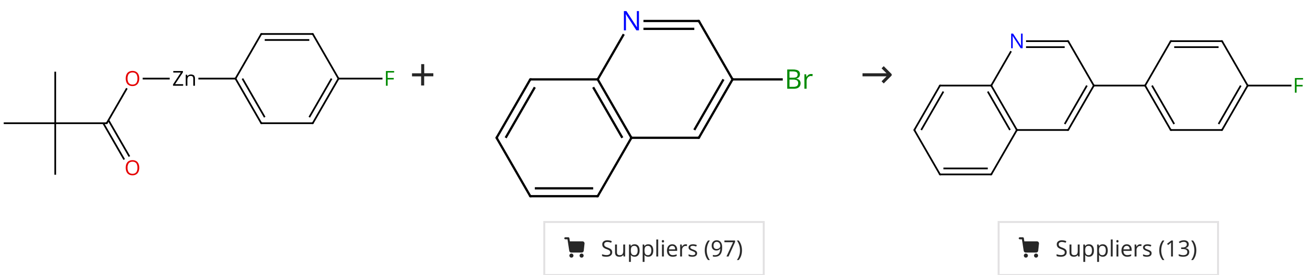
Mild Negishi Cross-Coupling Reactions Catalyzed by Acenaphthoimidazolylidene Palladium Complexes at Low Catalyst Loadings

By: Liu, Zelong; et al

Journal of Organic Chemistry (2013), 78(15), 7436-7444.

Scheme 30 (1 Reaction)

Steps: **1** Yield: **99%**



31-176-CAS-1603558

Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Tetrahydrofuran; 1 h, 25 °C

Experimental Protocols

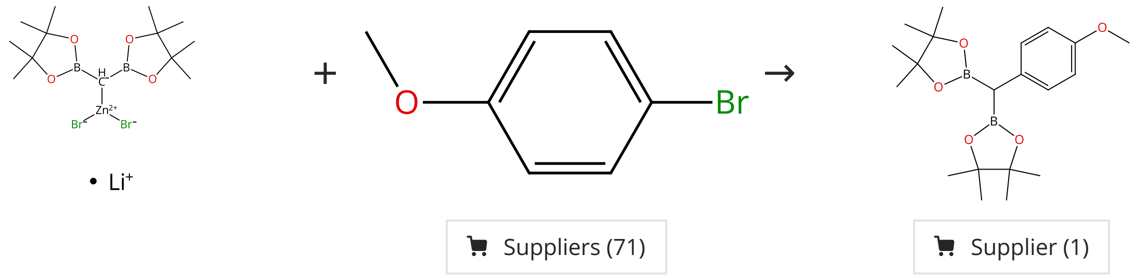
Preparation of solid salt-stabilized functionalized organozinc compounds and their application to cross-coupling and carbonyl addition reactions

By: Bernhardt, Sebastian; et al

Angewandte Chemie, International Edition (2011), 50(39), 9205-9209, S9205/1-S9205/68.

Scheme 31 (1 Reaction)

Steps: **1** Yield: **99%**



31-176-CAS-20426570	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Tri-*o*-tolylphosphine, Tris(dibenzylideneacetone) dipalladium **Solvents:** Tetrahydrofuran; 3 h, 60 °C

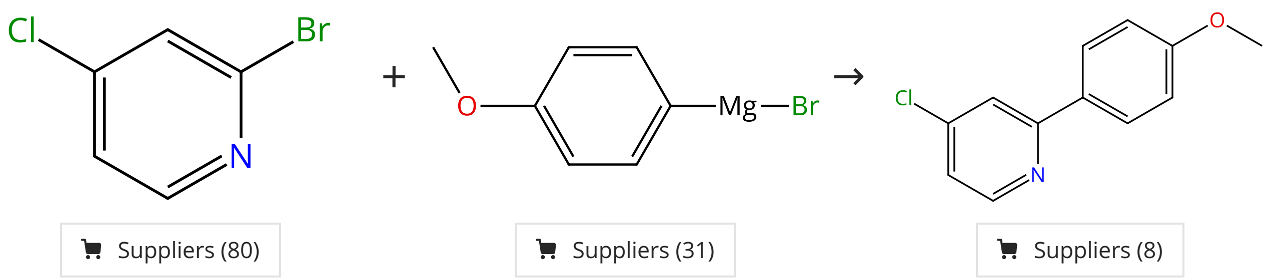
Palladium-Catalyzed Chemoselective Negishi Cross-Coupling of Bis[(pinacolato)boryl]methylzinc Halides with Aryl (Pseudo) Halides

By: Lee, Hyojae; et al

Organic Letters (2019), 21(15), 5912-5916.

Scheme 32 (1 Reaction)

Steps: **1** Yield: **99%**



31-614-CAS-32972820	Steps: **1** Yield: **99%**
1.1	**Catalysts:** Palladium diacetate, 1,1-Bis(diphenylphosphino) ferrocene **Solvents:** Tetrahydrofuran; 2 h, 25 °C

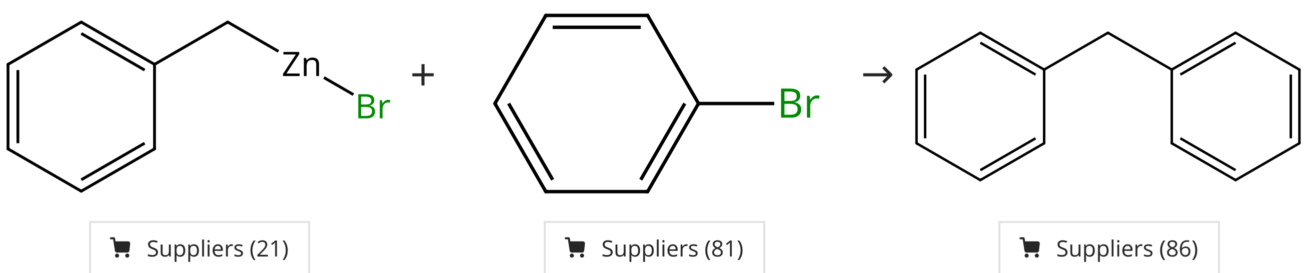
Unconventional Site Selectivity in Palladium-Catalyzed Cross-Couplings of Dichloroheteroarenes under Ligand-Controlled and Ligand-Free Systems

By: Norman, Jacob P.; et al

Journal of Organic Chemistry (2022), 87(11), 7414-7421.

Scheme 33 (1 Reaction)

Steps: **1** Yield: **99%**



31-176-CAS-1445176	Steps: **1** Yield: **99%**
1.1	**Catalysts:** (*SP*-4-4)-[(Benzoic acid-κ*O*) 2-[3,3,3-trifluoro-3-(oxo-κ*O*)-1-(trifluoromethyl)butyl]hydrazidato(2-)-κN^2](triphenyl phosphine)nickel **Solvents:** Tetrahydrofuran; 24 h, 70 °C

Experimental Protocols

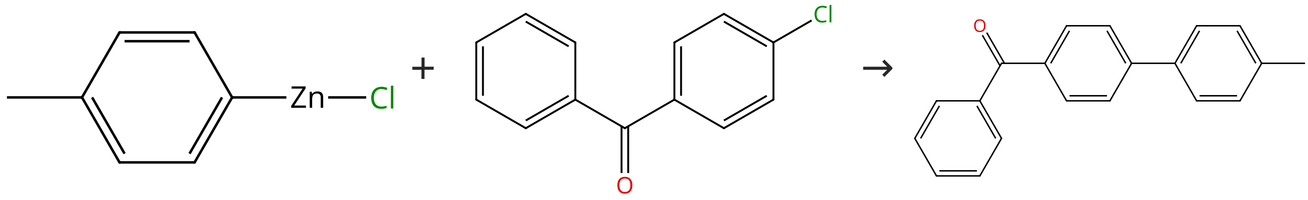
Synthesis of Ni(II) complexes with unsymmetric [O,N,O']-pincer ligands and their use as precatalysts in carbon-carbon bond formations to access diarylmethanes

By: Someya, Chika I.; et al

Inorganica Chimica Acta (2014), 421, 136-144.

Scheme 34 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (2)

Suppliers (92)

Suppliers (18)

31-176-CAS-6462746

Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-4)-Chloro[*N*2-[2-(diphenylphosphino-κ*P*)phenyl]-*N*1,*N*1-dimethyl-1,2-benzenediaminato-κ*N*1,κ*N*2]nickel
Solvents: Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; 24 h, 80 °C

Experimental Protocols

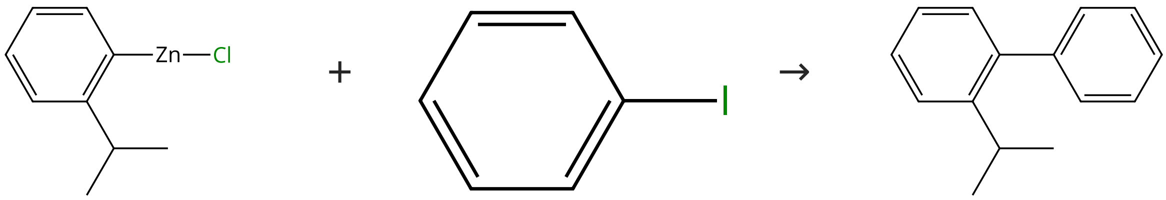
P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides

By: Wu, Dan; et al

Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.

Scheme 35 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (95)

Suppliers (14)

31-176-CAS-6301858

Steps: **1** Yield: **99%**

1.1 **Catalysts:** [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium
Solvents: Tetrahydrofuran; 2 h, 60 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

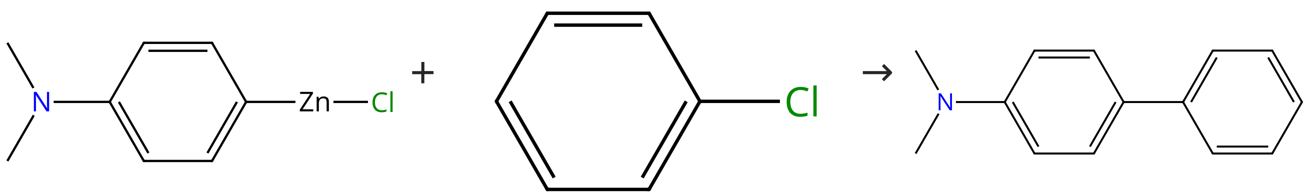
Revealing a Second Transmetalation Step in the Negishi Coupling and Its Competition with Reductive Elimination: Improvement in the Interpretation of the Mechanism of Biaryl Syntheses

By: Liu, Qiang; et al

Journal of the American Chemical Society (2009), 131(29), 10201-10210.

Scheme 36 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (3)

Suppliers (127)

Suppliers (23)

31-176-CAS-10503150

Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-4)-Chloro[*N*2-[2-(diphenylphosphino-κ*P*)phenyl]-*N*1,*N*1-dimethyl-1,2-benzenediaminato-κ*N*1,κ*N*2]nickel
Solvents: Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; 24 h, 80 °C

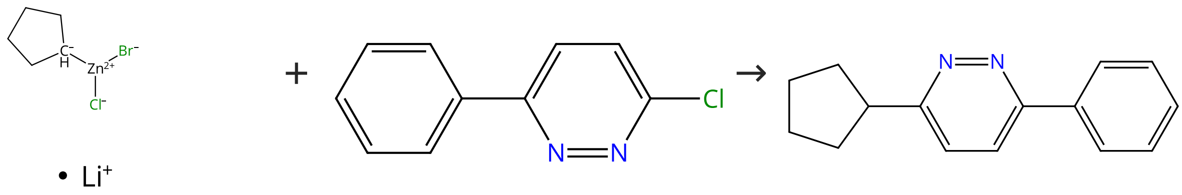
Experimental Protocols

P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides

By: Wu, Dan; et al

Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.

Scheme 37 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (77)

Suppliers (2)

31-176-CAS-2359273 Steps: **1** Yield: **99%**

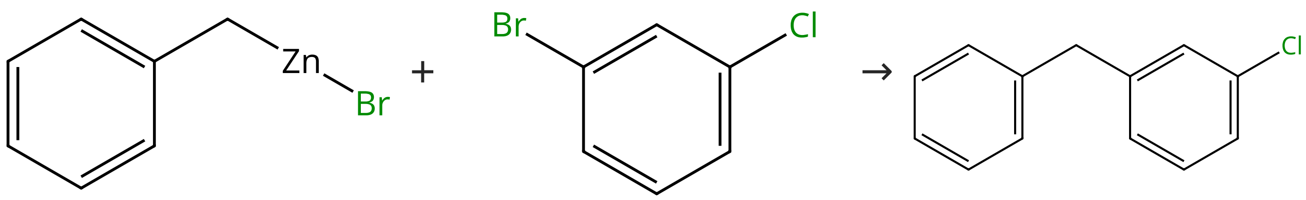
1.1 **Catalysts:** (*SP*-4-1)-[7,9-Bis[2,6-bis(1-methylethyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: 1,4-Dioxane; 0 °C; 6 h, rt

Mild Negishi Cross-Coupling Reactions Catalyzed by Acenaphthoimidazolylidene Palladium Complexes at Low Catalyst Loadings

By: Liu, Zelong; et al

Journal of Organic Chemistry (2013), 78(15), 7436-7444.

Scheme 38 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (21)

Suppliers (82)

Suppliers (9)

31-176-CAS-20748296 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-3)-Bromophenylbis(triphenylphosphine)palladium
Solvents: Tetrahydrofuran; rt; 6 h, 50 °C

1.2 **Reagents:** Ammonium chloride
Solvents: Water

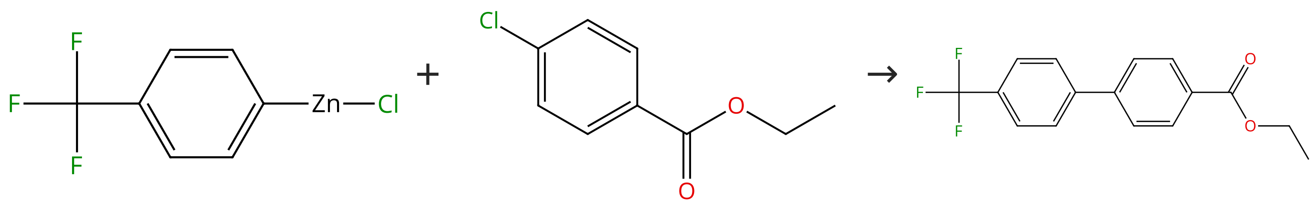
Experimental Protocols

POxAP Precatalysts and the Negishi Cross-Coupling Reaction

By: Tang, Shuang-Qi; et al

Synthesis (2020), 52(1), 51-59.

Scheme 39 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (2)

Suppliers (71)

Suppliers (13)

31-176-CAS-13613817 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-4)-Chloro[N^2-[2-(diphenylphosphino-κ*P*)phenyl]-N^1,N^1-dimethyl-1,2-benzenediaminato-κN^1,κN^2]nickel
Solvents: Tetrahydrofuran, *N*-Methyl-2-pyrrolidone; 24 h, 80 °C

Experimental Protocols

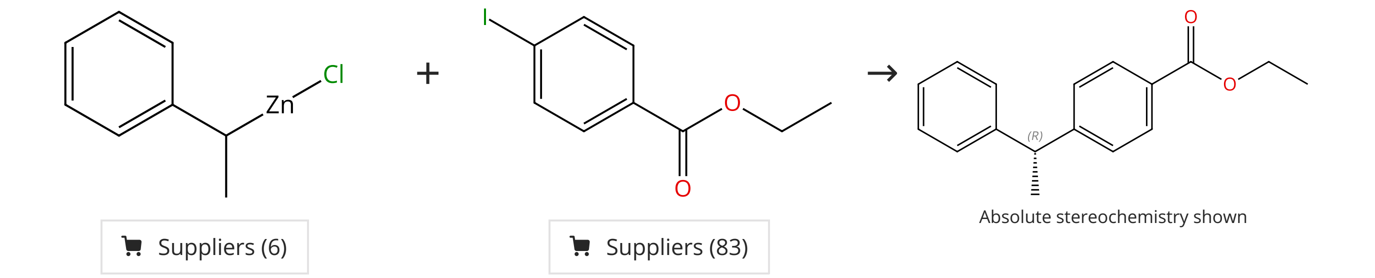
P,N,N-Pincer nickel-catalyzed cross-coupling of aryl fluorides and chlorides

By: Wu, Dan; et al

Organic & Biomolecular Chemistry (2014), 12(33), 6414-6424.

Scheme 40 (1 Reaction)

Steps: **1** Yield: **99%**



31-614-CAS-42728041

Steps: **1** Yield: **99%**

1.1 **Catalysts:** Di-μ-chlorobis[(1,2,3-η)-1-phenyl-2-propen-1-yl]dipalladium, 1,3-Dioxolo[4,5-*e*][1,3,2]dioxaphosphepin, 6-[[(1*R*,2*S*,5*R*)-2-[1-[3,5-bis(1,1-dimethylethyl)-4-methoxyphenyl]-1-methylethyl]-5-methylcyclohexyl]oxy]-4,4,8,8-tetrakis[4-(1,1-dimethylethyl)phenyl]tetrahydro-2,2-dimethyl-, (3a*R*,8a*R*)-
Solvents: *tert*-Butyl methyl ether; 30 min, rt

1.2 **Reagents:** Zinc chloride
Solvents: 2-Methyltetrahydrofuran, *tert*-Butyl methyl ether; 12 h, 21 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

Experimental Protocols

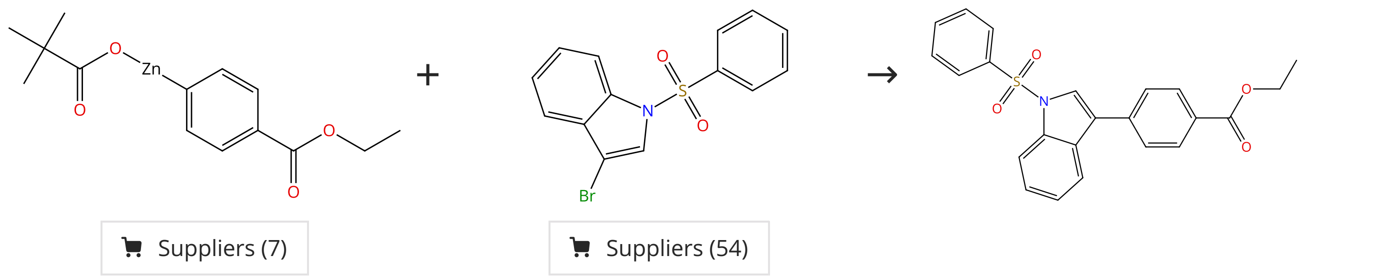
Enantioconvergent Negishi Cross-Couplings of Racemic Secondary Organozinc Reagents to Access Privileged Scaffolds: A Combined Experimental and Theoretical Study

By: Preinfalk, Alexander; et al

Angewandte Chemie, International Edition (2024), 63(50), e202414868.

Scheme 41 (1 Reaction)

Steps: **1** Yield: **99%**



31-176-CAS-13114457

Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*)palladium
Solvents: Ethyl acetate; 1 h, 25 °C

Experimental Protocols

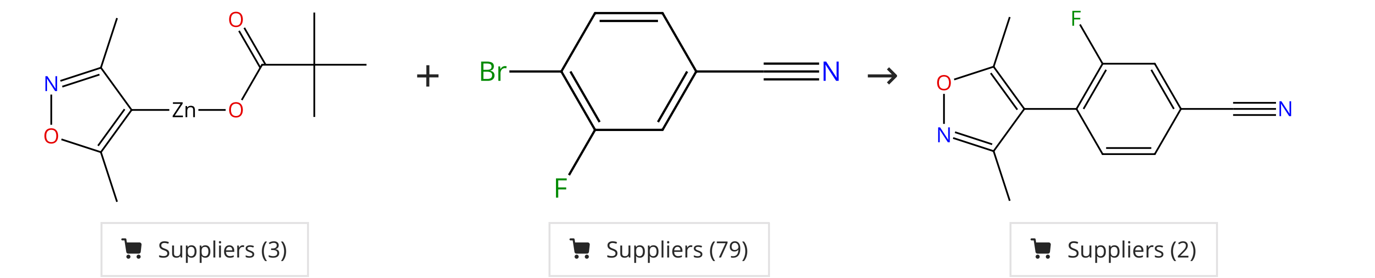
Preparation of solid salt-stabilized functionalized organozinc compounds and their application to cross-coupling and carbonyl addition reactions

By: Bernhardt, Sebastian; et al

Angewandte Chemie, International Edition (2011), 50(39), 9205-9209, S9205/1-S9205/68.

Scheme 42 (1 Reaction)

Steps: **1** Yield: **99%**



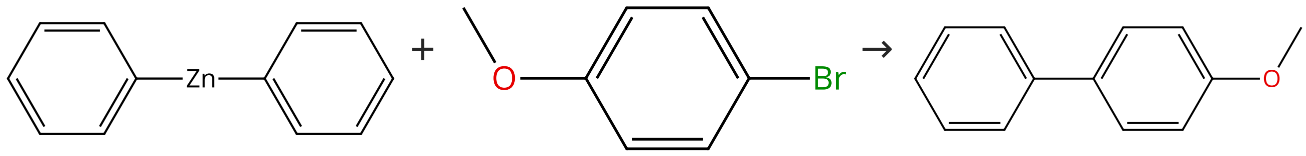
31-176-CAS-8516441	Steps: **1** Yield: **99%**
1.1	**Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-methylethyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*) palladium **Solvents:** Tetrahydrofuran; 2 h, 50 °C
Experimental Protocols	

Preparation of solid salt-stabilized functionalized organozinc compounds and their application to cross-coupling and carbonyl addition reactions

By: Bernhardt, Sebastian; et al

Angewandte Chemie, International Edition (2011), 50(39), 9205-9209, S9205/1-S9205/68.

Scheme 43 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (31) Suppliers (71) Suppliers (69)

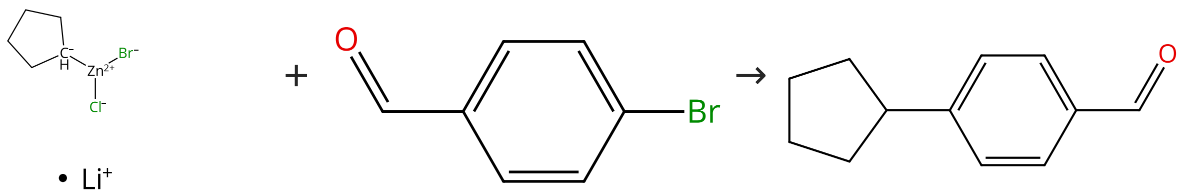
31-176-CAS-14600917	Steps: **1** Yield: **99%**
1.1	**Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*) palladium **Solvents:** Tetrahydrofuran; 2 h, rt

On the remarkably different role of salt in the cross-coupling of arylzincs from that seen with alkylzincs

By: McCann, Lucas C.; et al

Angewandte Chemie, International Edition (2014), 53(17), 4386-4389.

Scheme 44 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (91) Suppliers (44)

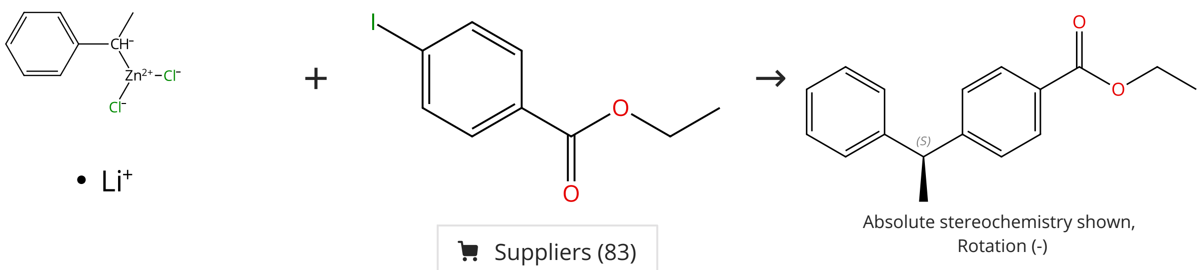
31-176-CAS-5131990	Steps: **1** Yield: **99%**
1.1	**Catalysts:** (*SP*-4-1)-[7,9-Bis[2,6-bis(1-methylethyl)phenyl]-7,9-dihydro-8*H*-acenaphth[1,2-*d*]imidazol-8-ylidene]dichloro(3-chloropyridine-κ*N*)palladium **Solvents:** 1,4-Dioxane; 0 °C; 0.5 h, rt

Mild Negishi Cross-Coupling Reactions Catalyzed by Acenaphthoimidazolylidene Palladium Complexes at Low Catalyst Loadings

By: Liu, Zelong; et al

Journal of Organic Chemistry (2013), 78(15), 7436-7444.

Scheme 45 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (83)

31-614-CAS-42728052 Steps: **1** Yield: **99%**

1.1 **Catalysts:** Palladium diacetate, 1,3-Dioxolo[4,5-*e*][1,3,2]dioxaphosphepin, 4,4,8,8-tetrakis[4-(1,1-dimethylethyl)phenyl]tetrahydro-2,2-dimethyl-6-[[(1*S*,2*R*,5*S*)-5-methyl-2-(1-methylethyl)cyclohexyl]oxy]-, (3a*S*,8a*S*)-
Solvents: *tert*-Butyl methyl ether; 30 min, rt

1.2 **Reagents:** Zinc chloride
Solvents: 2-Methyltetrahydrofuran, *tert*-Butyl methyl ether; 12 h, 23 °C

1.3 **Reagents:** Hydrochloric acid
Solvents: Water

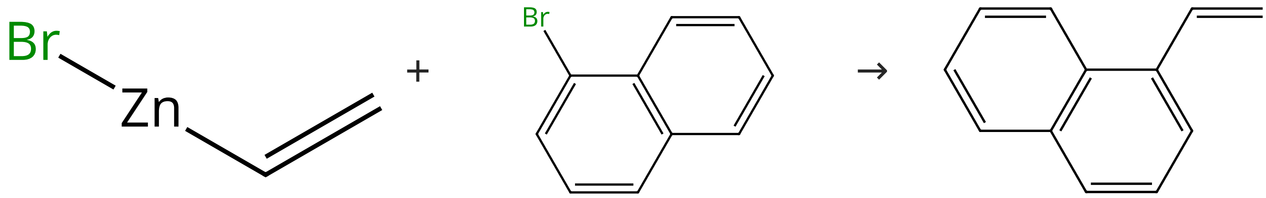
Experimental Protocols

Enantioconvergent Negishi Cross-Couplings of Racemic Secondary Organozinc Reagents to Access Privileged Scaffolds: A Combined Experimental and Theoretical Study

By: Preinfalk, Alexander; et al

Angewandte Chemie, International Edition (2024), 63(50), e202414868.

Scheme 46 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (9) Suppliers (91) Suppliers (67)

31-176-CAS-14773885 Steps: **1** Yield: **99%**

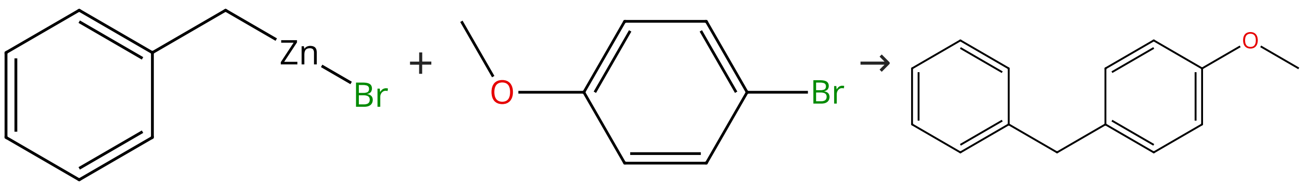
1.1 **Catalysts:** 1637333-83-5
Solvents: Tetrahydrofuran; 5 h, 50 °C

Synthesis, mechanism of formation, and catalytic activity of Xantphos nickel π-complexes

By: Staudaher, Nicholas D.; et al

Chemical Communications (Cambridge, United Kingdom) (2014), 50(98), 15577-15580.

Scheme 47 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (21) Suppliers (71) Suppliers (53)

31-176-CAS-10302652 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-4)-[(Benzoic acid-κ*O*) 2-[3,3,3-trifluoro-3-(oxo-κ*O*)-1-(trifluoromethyl)butyl]hydrazidato(2-)-κN^2](triphenyl phosphine)nickel
Solvents: Tetrahydrofuran; 24 h, 70 °C

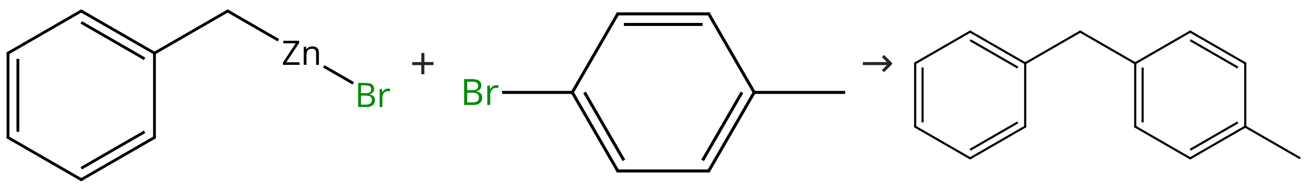
Experimental Protocols

Synthesis of Ni(II) complexes with unsymmetric [O,N,O']-pincer ligands and their use as precatalysts in carbon-carbon bond formations to access diarylmethanes

By: Someya, Chika I.; et al

Inorganica Chimica Acta (2014), 421, 136-144.

Scheme 48 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (21) Suppliers (67) Suppliers (61)

31-176-CAS-12392434 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-4)-[(Benzoic acid-κ*O*) 2-[3,3,3-trifluoro-3-(oxo-κ*O*)-1-(trifluoromethyl)butyl]hydrazidato(2-)-κN^2](triphenyl phosphine)nickel
Solvents: Tetrahydrofuran; 24 h, 70 °C

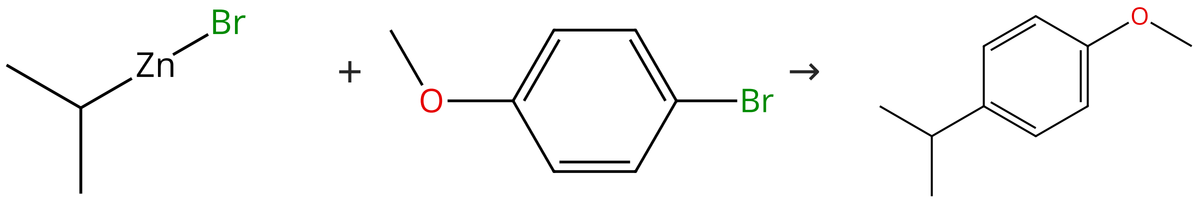
Experimental Protocols

Synthesis of Ni(II) complexes with unsymmetric [O,N,O']-pincer ligands and their use as precatalysts in carbon-carbon bond formations to access diarylmethanes

By: Someya, Chika I.; et al

Inorganica Chimica Acta (2014), 421, 136-144.

Scheme 49 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (21) Suppliers (71) Suppliers (63)

31-176-CAS-4961267 Steps: **1** Yield: **99%**

1.1 **Catalysts:** (*SP*-4-1)-[1,3-Bis[2,6-bis(1-ethylpropyl)phenyl]-1,3-dihydro-2*H*-imidazol-2-ylidene]dichloro(3-chloropyridine-κ*N*) palladium
Solvents: Toluene, Tetrahydrofuran; 2 min, 0 °C; 0 °C → rt; 30 min, rt

1.2 **Reagents:** Hydrochloric acid
Solvents: Water; rt

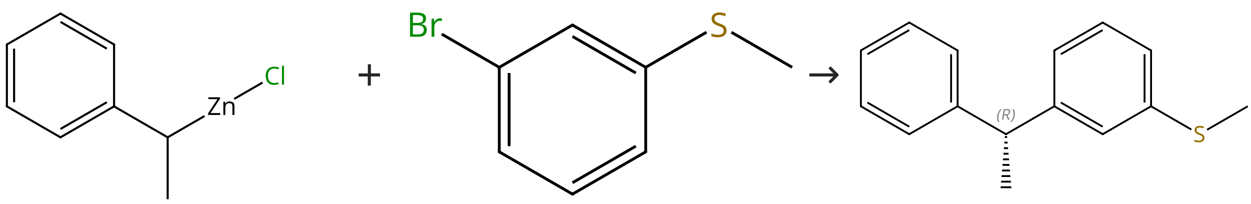
Experimental Protocols

Negishi cross-coupling of secondary alkylzinc halides with aryl/heteroaryl halides using Pd-PEPPSI-IPent

By: Calimsiz, Selcuk; et al

Chemical Communications (Cambridge, United Kingdom) (2011), 47(18), 5181-5183.

Scheme 50 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (6) Suppliers (75)

31-614-CAS-42728024 Steps: **1** Yield: **99%**	**Enantioconvergent Negishi Cross-Couplings of Racemic Secondary Organozinc Reagents to Access Privileged Scaffolds: A Combined Experimental and Theoretical Study**
1.1 **Catalysts:** Di-μ-chlorobis[(1,2,3-η)-1-phenyl-2-propen-1-yl]dipalladium, 1,3-Dioxolo[4,5-*e*][1,3,2]dioxaphosphepin, 4,4,8,8-tetrakis[4-(1,1-dimethylethyl)phenyl]tetrahydro-2,2-dimethyl-6-[[(1*R*,2*S*,5*R*)-5-methyl-2-(1-methyl-1-phenylethyl)cyclohexyl]oxy]-, (3a*R*,8a*R*)- **Solvents:** *tert*-Butyl methyl ether; 30 min, rt 1.2 **Reagents:** Zinc chloride **Solvents:** 2-Methyltetrahydrofuran, *tert*-Butyl methyl ether; 12 h, 21 °C 1.3 **Reagents:** Hydrochloric acid **Solvents:** Water	By: Preinfalk, Alexander; et al Angewandte Chemie, International Edition (2024), 63(50), e202414868.
Experimental Protocols	